

**Notation used throughout**

| Symbol                     | Meaning                                                                              |
|----------------------------|--------------------------------------------------------------------------------------|
| $\Omega$                   | sample space                                                                         |
| $A, B$                     | events                                                                               |
| $A^c$                      | complement of $A$                                                                    |
| $X, Y$                     | random variables                                                                     |
| $x_i, y_i$                 | observed values in row $i$                                                           |
| $n_A, n, N$                | count in $A$ , sample size, population size                                          |
| $\mu, \sigma^2, \sigma, p$ | population mean, variance, standard deviation, proportion                            |
| $\bar{X}, S^2, \hat{P}$    | estimators: sample mean, variance, proportion                                        |
| $\bar{x}, s^2, \hat{p}$    | estimates: their values in one sample                                                |
| $\mu_0, p_0$               | values of $\mu$ and $p$ under $H_0$                                                  |
| $\hat{\beta}_j, \beta_j$   | estimated coefficient (with a hat), population coefficient                           |
| $SD(\bar{X}), SE(\bar{X})$ | standard deviation of an estimator, and standard error: its estimate from the sample |
| $SD_0(\hat{P})$            | standard deviation of $\hat{P}$ under $H_0$                                          |
| $z_q, t_{q, df}$           | $q$ quantile of $N(0, 1)$ and of $t_{df}$ : $P(Z \leq z_q) = q$                      |
| $z_{obs}, t_{obs}$         | observed value of a test statistic                                                   |
| $\alpha$                   | significance level                                                                   |
| $df$                       | degrees of freedom                                                                   |

**Lecture 1 - Introduction to data analysis****Proportions and percentages**

$$\text{proportion} = \frac{n_A}{n}, \quad \text{percentage} = 100 \frac{n_A}{n} \%.$$

**Sample mean and range**

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i, \quad \text{range} = \max_i x_i - \min_i x_i.$$

**Median, quartiles, and interquartile range**

$Q_1$ , median,  $Q_3$ : values with at least 25, 50, and 75 percent of the observations at or below them, and at least 75, 50, and 25 percent at or above them.

$$\text{IQR} = Q_3 - Q_1.$$

**Lecture 2 - Probability foundations and random variables****Probability bounds, sample space, and complement**

$$0 \leq P(A) \leq 1, \quad P(\Omega) = 1, \quad P(A^c) = 1 - P(A).$$

**Addition rule**

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

**Addition rule for disjoint events**

$$A \cap B = \emptyset \implies P(A \cup B) = P(A) + P(B).$$

**Equally likely outcomes**

$$P(A) = \frac{|A|}{|\Omega|} \quad (\text{equally likely outcomes}).$$

**Three more properties**

$$P(\emptyset) = 0, \quad A \subseteq B \implies P(A) \leq P(B), \\ P(A) = P(A \cap B) + P(A \cap B^c).$$

**Partition of the sample space**

$$B_i \cap B_j = \emptyset \quad (i \neq j), \quad \bigcup_{i=1}^k B_i = \Omega.$$

**Lecture 3 - Discrete distributions, expectation, and variance****Probability mass function (PMF)**

$$p_X(x) = P(X = x), \quad p_X(x) \geq 0, \quad \sum_{x \in \mathcal{X}} p_X(x) = 1.$$

**Cumulative distribution function (CDF)**

$$F_X(a) = P(X \leq a) = \sum_{x \leq a} p_X(x).$$

**Interval probability from a CDF**

$$P(a < X \leq b) = F_X(b) - F_X(a).$$

**Expectation and expectation of a transformation**

$$E[X] = \sum_{x \in \mathcal{X}} x p_X(x), \quad E[g(X)] = \sum_{x \in \mathcal{X}} g(x) p_X(x).$$

**Population variance and standard deviation**

$$\mu = E[X], \quad \sigma^2 = \text{Var}(X) = E[(X - \mu)^2] = E[X^2] - (E[X])^2, \\ \sigma = \text{SD}(X) = \sqrt{\text{Var}(X)}.$$

**Shifting and rescaling a variable**

$$E[aX + b] = aE[X] + b, \quad \text{Var}(aX + b) = a^2 \text{Var}(X), \\ \text{SD}(aX + b) = |a| \text{SD}(X).$$

**Bernoulli distribution**

$$X \sim \text{Bernoulli}(p): \quad P(X = 1) = p, \quad P(X = 0) = 1 - p, \\ E[X] = p, \quad \text{Var}(X) = p(1 - p).$$

**Sample variance and standard deviation**

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2, \quad s = \sqrt{s^2}.$$

**Lecture 4 - Conditioning, Bayes' rule, and independence****Conditional probability and multiplication rule**

$$P(A | B) = \frac{P(A \cap B)}{P(B)}, \quad P(A \cap B) = P(A | B)P(B), \\ P(B) > 0.$$

**Law of total probability**

$$P(A) = \sum_{i=1}^k P(A | B_i)P(B_i).$$

**Bayes' rule**

$$P(B_i | A) = \frac{P(A | B_i)P(B_i)}{\sum_{j=1}^k P(A | B_j)P(B_j)}.$$

**Bayes' rule for one event  $B$** 

$$P(B | A) = \frac{P(A | B)P(B)}{P(A)}, \quad P(A^c | B) = 1 - P(A | B).$$

**Independence of events**

$$A \text{ and } B \text{ independent} \iff P(A \cap B) = P(A)P(B).$$

**Lecture 5 - Joint random variables, covariance, and correlation****Joint PMF**

$$p_{X,Y}(x, y) = P(X = x, Y = y).$$

**Marginal and conditional PMFs**

$$p_X(x) = \sum_y p_{X,Y}(x, y), \quad p_{X|Y}(x | y) = \frac{p_{X,Y}(x, y)}{p_Y(y)}.$$

**Conditional mean**

$$E[X | Y = y] = \sum_x x p_{X|Y}(x | y).$$

**Independence of random variables**

$$X \text{ and } Y \text{ independent} \iff p_{X,Y}(x, y) = p_X(x)p_Y(y) \\ \text{for every } x, y.$$

**Population covariance and correlation**

$$\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)] = E[XY] - E[X]E[Y].$$

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\text{SD}(X)\text{SD}(Y)}, \quad -1 \leq \rho_{XY} \leq 1.$$

**Sample covariance and correlation**

$$s_{XY} = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}), \quad r_{XY} = \frac{s_{XY}}{s_X s_Y}.$$

**Sums of two variables**

$$E[X + Y] = E[X] + E[Y],$$

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y).$$

$$X \text{ and } Y \text{ independent} \implies$$

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y).$$

**Lecture 6 - Continuous distributions and the normal model****Probabilities for a continuous distribution**

$$f_X(x) \geq 0, \quad \text{total area under } f_X = 1, \quad P(X = x) = 0,$$

$$P(a < X \leq b) = F_X(b) - F_X(a).$$

**Uniform distribution**

$$X \sim U(a, b): \quad f_X(x) = \begin{cases} \frac{1}{b-a}, & a \leq x \leq b, \\ 0, & \text{otherwise,} \end{cases}$$

$$P(c < X < d) = \frac{d-c}{b-a}, \quad E[X] = \frac{a+b}{2},$$

$$\text{Var}(X) = \frac{(b-a)^2}{12}, \quad \text{SD}(X) = \frac{b-a}{\sqrt{12}}.$$

**Normal distribution and standardisation**

$$X \sim N(\mu, \sigma^2), \quad Z = \frac{X - \mu}{\sigma} \sim N(0, 1).$$

$$P(Z > z) = 1 - P(Z \leq z), \quad P(Z \leq -z) = P(Z > z),$$

$$x = \mu + z\sigma.$$

**Quantiles and the central part of the standard normal distribution**

$$P(X \leq x_q) = q, \quad x_q = \mu + z_q\sigma,$$

$$P(-z_{1-\alpha/2} \leq Z \leq z_{1-\alpha/2}) = 1 - \alpha.$$

**Lecture 7 - Samples, parameters, and estimators****Sample mean and sample proportion estimators**

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i, \quad \hat{P} = \frac{1}{n} \sum_{i=1}^n X_i \quad (X_i \in \{0, 1\}).$$

**Sample variance and degrees of freedom**

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2, \quad \sum_{i=1}^n (X_i - \bar{X}) = 0, \quad df = n - 1.$$

**Shortcut for the sample variance, and the variance of a 0/1 variable**

$$S^2 = \frac{\sum_{i=1}^n X_i^2 - n\bar{X}^2}{n-1},$$

$$X_i \in \{0, 1\}: \quad \sigma^2 = p(1-p), \quad \text{estimate } \hat{p}(1-\hat{p}).$$

**Unbiased estimators**

$$E[\bar{X}] = \mu, \quad E[\hat{P}] = p, \quad E[S^2] = \sigma^2.$$

**Random sample, and the 10% condition for sampling without replacement**

$$E[X_i] = \mu, \quad \text{Var}(X_i) = \sigma^2,$$

$$n \leq 0.10N \implies X_1, \dots, X_n \text{ treated as iid.}$$

**Lecture 8 - Sampling distributions and the central limit theorem****Sample mean: standard deviation and standard error**

$$E[\bar{X}] = \mu, \quad \text{Var}(\bar{X}) = \frac{\sigma^2}{n}, \quad \text{SD}(\bar{X}) = \frac{\sigma}{\sqrt{n}}, \quad \text{SE}(\bar{X}) = \frac{\sigma}{\sqrt{n}}.$$

**Distribution of the sample mean (exact for a normal population, approximate for large  $n$ )**

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right), \quad \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1).$$

**Sample proportion: standard deviation and standard error**

$$E[\hat{P}] = p, \quad \text{SD}(\hat{P}) = \sqrt{\frac{p(1-p)}{n}}, \quad \text{SE}(\hat{P}) = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}.$$

**Number of successes in  $n$  independent trials (binomial)**

$$Y \sim \text{Bin}(n, p): \quad E[Y] = np, \quad \text{Var}(Y) = np(1-p),$$

$$\hat{P} = \frac{Y}{n}.$$

**Success-failure condition**

$$np \geq 10, \quad n(1-p) \geq 10.$$

**Lecture 9 - The t distribution and confidence intervals****One-sample  $t$  statistic**

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}$$

(normal population, or approximately for large  $n$ ).

**Confidence interval for a mean and for a proportion (level  $1 - \alpha$ )**

$$\bar{X} \pm t_{1-\alpha/2, n-1} \frac{S}{\sqrt{n}},$$

$$\hat{P} \pm z_{1-\alpha/2} \sqrt{\frac{\hat{P}(1-\hat{P})}{n}} \quad (n\hat{p} \geq 10, \quad n(1-\hat{p}) \geq 10).$$

**Critical values  $z_{1-\alpha/2}$ , and  $t_{1-\alpha/2, df}$  for  $df = 5$  to 120**

| $1 - \alpha$ | $z_{1-\alpha/2}$ | $df=5$ | 10    | 20    | 30    | 60    | 100   | 120   |
|--------------|------------------|--------|-------|-------|-------|-------|-------|-------|
| 0.90         | 1.645            | 2.015  | 1.812 | 1.725 | 1.697 | 1.671 | 1.660 | 1.658 |
| 0.95         | 1.960            | 2.571  | 2.228 | 2.086 | 2.042 | 2.000 | 1.984 | 1.980 |
| 0.99         | 2.576            | 4.032  | 3.169 | 2.845 | 2.750 | 2.660 | 2.626 | 2.617 |

If your  $df$  is not listed, use the closest listed value.

**Sample size for a margin of error  $m$  around a mean (round up)**

$$n \geq \left( \frac{z_{1-\alpha/2} \sigma}{m} \right)^2.$$

**Confidence interval in five parts**

1 formula, 2 given values, 3 calculations (standard error, margin of error), 4 interval, 5 conclusion.

**Lecture 10 - Testing decisions, errors, power, and a first look at regression****Test in five steps**

1 hypotheses and  $\alpha$ , 2 test statistic and its distribution under  $H_0$ , 3 critical value and rejection region, 4 observed value, 5 conclusion.

**Two-sided test for a mean**

$$H_0: \mu = \mu_0 \quad \text{versus} \quad H_1: \mu \neq \mu_0,$$

$$T = \frac{\bar{X} - \mu_0}{S/\sqrt{n}} \sim t_{n-1} \text{ under } H_0,$$

$$\text{reject } H_0 \text{ if } |t_{obs}| \geq t_{1-\alpha/2, n-1}.$$

**Two-sided test for a proportion ( $np_0 \geq 10, n(1-p_0) \geq 10$ )**

$$H_0: p = p_0 \quad \text{versus} \quad H_1: p \neq p_0,$$

$$Z = \frac{\hat{P} - p_0}{\text{SD}_0(\hat{P})} \sim N(0, 1) \text{ under } H_0, \quad \text{SD}_0(\hat{P}) = \sqrt{\frac{p_0(1-p_0)}{n}},$$

$$\text{reject } H_0 \text{ if } |z_{obs}| \geq z_{1-\alpha/2}.$$

**Two-sided  $p$ -value**

$$p = 2P(T \geq |t_{obs}|) \quad \text{or} \quad 2P(Z \geq |z_{obs}|),$$

$$p \leq \alpha \iff \text{reject } H_0.$$

**Test and  $(1 - \alpha)$  confidence interval for a mean**

$$\text{reject } H_0: \mu = \mu_0 \text{ at level } \alpha \iff \mu_0 \text{ outside the interval.}$$

**Type I error, Type II error, and power**

$$P(\text{Type I error}) \leq \alpha, \quad \text{Power} = 1 - P(\text{Type II error}).$$

**Regression coefficient ( $H_0: \beta_j = 0$  versus  $H_1: \beta_j \neq 0$ )**

$$t_{obs} = \hat{\beta}_j / \text{SE}(\hat{\beta}_j), \quad \hat{\beta}_j \pm t_{1-\alpha/2, df} \text{SE}(\hat{\beta}_j).$$